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ORIGINAL PAPER

YOLO-based anomaly activity detection system for human behavior analysis and crime mitigation

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Abstract
Detecting the objects and tracking them is a wonder with the help of the recent technologies. In today's life, video surveillance is commonly present at maximum places. As the population increases, the crime rates also keep on increasing at different cases. The crime identification, dealing with the crime scenes and tracking down the criminal is a major task which involves numerous man powers. If there is a methodology for detecting and preventing the crime, it would be very much helpful to the public and also to the authorities. So the main objective of the proposed system is to automate the crime identification and crime tracking process by using the video surveillance with deep learning algorithms. Also, to alert the public before a crime takes place in public areas. Because of the timely alert, further crime activities may be avoided or the loss incurred may be reduced. The system also helps in identifying the criminals and tracking the information about the crime scenes, thus providing more useful information to the authorities in solving the cases. According to recent research works, the YOLO algorithm achieves higher accuracy with multi-object detection. In this paper, a framework with improvised YOLO algorithm is illustrated. The algorithm is fine-tuned with different hyperparameters for achieving the AUC with 0.91 for detecting vandalism behavior and 0.8299 over all the 14 classes of crime activities. The result of the proposed system is compared with the existing systems with parameters like training loss, testing loss, precision and F1 score.

Keywords Anomaly detection · Abnormal behavior · Anomaly activities · Yolo · Hyperparameters · Multi object detection

1 Introduction

In the last decade, the crime activities in India are keep on increasing which lies in the range of 422.2 per 1,00,000 people in our country. Most of the activities took place in public areas which includes murder, theft, physical abuse, burglary activities, etc. Violent activities against women and children like abuse and sexual assault have been kept on increasing day by day. These types of activities may be prevented/avoided if suitable alerting systems are deployed in the vulnerable places.

Nowadays CCTV has become one of the major components that are available in all public and private areas. Because of the increasing threats and violence activities happening in the society, there is a need for an automated system which will detect and report the abnormal events that are happening in an environment. For developing such autonomous systems, the CCTV surveillance cameras can be used. Human's activity and their behavior can be monitored with the help of CCTV's and these data can be used for developing an intelligent anomaly detection system which ensures the safety of people in a context.

In most cases, public places are crowded with high population which makes it difficult to monitor the environment and secure it with man power. Even if, man power is employed in such places, it is not possible to pay attention to each and every person for monitoring their behavior. This may result in occurrence of abnormal events like accidents, violent activities and other criminal activities. So instead of using man power, a trained intelligent system can be employed in such places which will monitor each and every person and analyze their behavior based on the trained pattern. Based on this analysis, major crime incidents can be prevented or they can

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be reported to the concerned authority for making immediate actions.

So, there is a need for an autonomous anomaly detection system [1-10] which monitors the environment and human beings for abnormal activities. Abnormal activities are activities that may vary from normal activities like walking, talking, running, sitting and lying. So the abnormal activities may be activities like hitting, pushing, fighting, falling, abusing, etc. These abnormal activities may lead to violent activities like attack, murder, kidnapping and other criminal activities. For preventing those types of situations, the anomaly detection system will monitor the environment, and in case of any abnormal activities like abuse, accident, fight, bullying, arrest, attack, burglary and murder [11], [12], it will report it to the management. The system must be able to classify the events as normal and abnormal events [13], [14] with the help of machine learning algorithms. Once there is possibility of any unusual activity, these abnormal events will be reported. Thus the automated anomaly detection system will identify and report the abnormal events in an effective way using object detection methodology and machine learning algorithm.

For implementing such anomaly detection systems, machine learning algorithms like CNN, ANN [15], RNN and LSTM-based algorithms are most widely used for detecting the human and also for recognizing their actions. Algorithms like GAN, decision tree, K-NN have been used for activity classification processes. So in this paper, the different algorithms used in existing systems will be studied and compared with the proposed system.

1.1 Anomaly detection system

Anomaly detection system is mainly used for monitoring and classifying human behavior based using some algorithms. A suitable anomaly detection system must be able to do two tasks. They are:

1. Extracting meaningful features from the dataset.
2. Decision-making strategy for activity classification.

In anomaly detection system, we will be using the video dataset as input. Extraction of meaningful and necessary features from the given dataset is one of the main tasks. The video features include features like trajectory features, Motion and appearance features. Most of the anomaly detection systems consider motion and appearance features for human behavior analysis. And for decision-making strategy, we can choose any machine learning algorithm like Naïve Bayes, Regression algorithms, SVM and Decision tree algorithms. If the system is capable of completing these two processes and achieves higher accuracy in predicting the abnormal events, then the system is said to be good anomaly detection system.

The rest of the paper is organized as follows: Reviews about the related works done in the field of anomaly detection system in Sect. 2. The next part Sect. 3 illustrates the proposed system with LSTM–CNN-based feature selection mechanism and improvised YOLO-based anomaly detection system. Section 4, describes the evaluation results of anomaly detection system. The final conclusions and future works have been showcased in Sect. 5 and 6.

2 Related works

Numerous researches are being carried out in video analytics for detection of anomaly activities that can be applicable in different domains like traffic monitoring, patient monitoring, security surveillance, human resource management, etc. The researchers mainly focus on extracting the features and classifying the events as normal and abnormal events.

Shubham Shinde et al., [1] developed a human activity recognition system with the help of YOLO object detection techniques. In this system, they have analyzed the human behavior in LIRIS video dataset. The dataset contains video sequence that describes the activities like discussion between people, entering room, leaving a room, telephonic conversation and handshaking events. Based on the label created during action detection and the relative confidence score for every five frames, they have predicted the type of activity a particular person is doing. In this paper, they have also proposed that even with a single frame in the video sequence it is possible to predict the human behavior. By deploying activity classification method with YOLO, they have achieved an accuracy of 88.35%.

Luo and Liu [2] proposed a sparse coding-based methodology for features selection along with recurrent neural network for activity recognition and activity classification [16]. The sparse coding technique was used for optimizing the parameters and selecting meaningful features. They have deployed this model in UCSD Pedestrian 1 dataset [17] which has been useful for outdoor anomaly detection. They have also applied the same algorithm on CUHK dataset which consists of videos with events like running, walking, throwing objects like that. They have achieved 81.71% accuracy for the CUHK dataset and 92.21% for the Pedestrian dataset [18].

Xu et al. [3] developed a machine learning-based human behavior analysis system. For detecting the motion features Optical flow is used. And the system uses CNN model for extracting the important features in the video dataset. For classification of human activities, they have applied SVM classifier. Weizmann and KTH datasets are used for the experimenting the algorithm and analyze their performance.

Overall accuracy of 93.5% and 89.9% is achieved by this system on Weizmann and KTH datasets, respectively.

Wang and Xia [4] proposed a deep learning-based anomaly detection system. For learning about the human behavior and extracting video features they employed a two-set stack de-noising autoencoder [19], [20]. This de-noising autoencoder will work similar to the functionality of autoencoders with the capability of accepting the corrupted input. They have also used agglomerative information bottleneck for dimension reduction of the selected input features. They have employed and tested these algorithms in two datasets such as BEHAVE [21] and BOSS dataset. Both the dataset comprises of video data for defining the normal events like walking, running and abnormal activities like fighting, chasing, harassing and panicking activities.

Xinfeng Zhang et al. [5] proposed a video anomaly detection and localization system for identifying abnormal behaviors. They have used optical flows between the consecutive frames for identifying the trajectory features in the video. They have applied a histogram-based shape description method for detection the trajectory information. This histogram-based shape descriptor is able to detect the abnormal activity even in narrow spaces with fast moving objects. They have also used a nonparametric K-NN algorithm for identifying the anomaly activities present in the system. They have achieved more than 90% AUC by applying this algorithm in the UMN dataset [22].

Guoqiang Liang et al. [6] proposed a unsupervised video summarization technique with convolutional attentive adversarial network which consists of a generator and discriminator module. For feature selection, LTSM is used. The generator in mentioned the system identifies the frame level important score, while the discriminator distinguishes the score weighted frame features from the original frame feature for summarizing the video. The model also track the long-term temporal information from the video for better summarization. The model analyzes impact of losses in the performance and reduced the loss level to less than 1.70%.

Vosta S et al. [7] proposed a system for detecting abnormal events from the camera feed. The have used RestNet50 for feature selection and ConvLSTM for detecting and classifying the abnormal events into 14 categories. The model used UCF-crime dataset and achieved an AUC of 81.71%.

Zin et al. [23] proposed a system for detecting unattended objects in public or private places with the help of consumer surveillance data.

Zin et al. [24] proposed a system for detecting loitering of people in public or private places using Markov random walk model (Table 1).

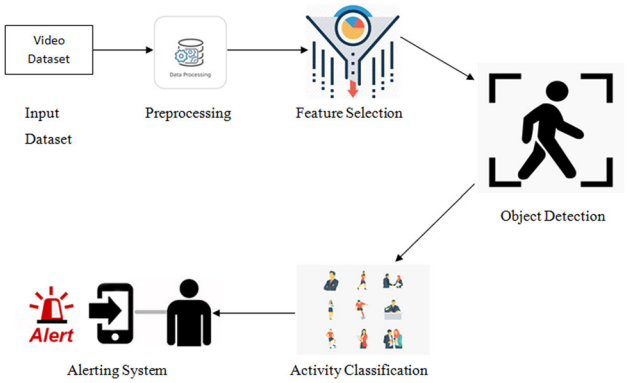


Fig. 1 System overview

3 Proposed system

In this paper, we propose an automated YOLO-based anomaly detection system, that is used for detecting and classifying the abnormal human behavior without human intervene. As per the discussion in the earlier Sect. 2, each anomaly detection system consists of two major components for extracting features and for decision-making regarding the activity classification.

An overview of the proposed system can be illustrated as follows (Fig. 1):

The anomaly detection system comprises of the following entities:

1. Dataset
2. Preprocessing
3. Feature selection and extraction
4. Object detection
5. Action classification

3.1 Dataset

The anomaly detection system takes the UCF- Crime dataset as input dataset. The dataset consists of video sequences that are captured at different places with different types of events. The events may be normal or abnormal events. So the video dataset is passed as input to the anomaly detection system. The video may have different properties and features like spatial features, appearance and motion features. Here in this video dataset we are going to consider the appearance and motion features for further processing. The appearance feature may include information regarding human posture, scene information, color and other information regarding the static objects in the environment. The motion features provide temporal information about the dataset. They may include features like motion speed, direction and movement of other components.

Table 1 Literature review is represented in Table 1

Author	Methodology	Dataset used	Performance measures achieved
Shubham Shinde et al [1]	YOLO—classified predefined normal activities	LIRIS video dataset	Achieved accuracy 88.35%
Luo and Liu [2]	Sparse coding for features selection along with recurrent neural network for activity recognition & classification	UCSD Pedestrian 1 dataset & CUHK dataset	81.71% accuracy for the CUHK dataset and 92.21% for pedestrian dataset
Xu et al [3]	CNN for feature selection & SVM for classification	Weizmann and KTH datasets	93.5% for Weizmann dataset and 89.9% for KTH datasets
Wang and Xia [4]	Two-set stack de-noising autoencoder	BEHAVE and BOSS dataset	92% accuracy achieved
Xinfeng Zhang et al [5]	Histogram-based shape description method for activity detection and KNN for classification	UMN dataset	Achieved 90% AUC
Guoqiang Liang et al [6]	LSTM for feature selection and convolutional attentive adversarial network (CAAN) for unsupervised video summarization	SumMe and TVSum dataset	Loss is reduced to 1.79%
Vosta et al. [7]	Abnormal event detection using RestNet50 and ConvLSTM technique	UCF-Crime dataset	Achieved 81.71 AUC
Zin et al [23]	Unattended object detection	Consumer surveillance data	Achieved more accurate predictions
Zin et al [23]	Markov model for loitering people detection	CCTV surveillance data	Achieved 95% accuracy

In this proposed system, we have used the UCF-Crime dataset which contains surveillance video footages that address 14 anomaly activities like abuse, fighting, assault, accidents, etc. This dataset also consists of video sequences of normal activities human which are captured in many indoor and outdoor areas. This video dataset comprises of 1900 video samples with an average frames of 7247 frames. Figure 2 depicts the different types of anomaly activities like assault, accidents and robbery activities.

The dataset consists of video sequences with normal events which may consider some sort of an anomaly event somewhere in the video. This will be recognized by the system and the anomaly events will be reported with the proposed algorithm. The distribution of the dataset over the 14 categories is represented in Figs. 3 and 4.

3.2 Preprocessing

The preprocessing process is also called as data cleaning. Before feeding the dataset to the algorithm for activity detection and recognition, it must be cleaned first. The videos cannot be fed into the system a whole sequence instead it will be divided into set of frames. The RGB color images in the frames will be grayscale images for easy processing. The motion features will be identified by finding the similarity in adjacent frames. As activity classification cannot be judged by a single frame, consecutive frames will be used for deciding the motion and the types of activity being played.

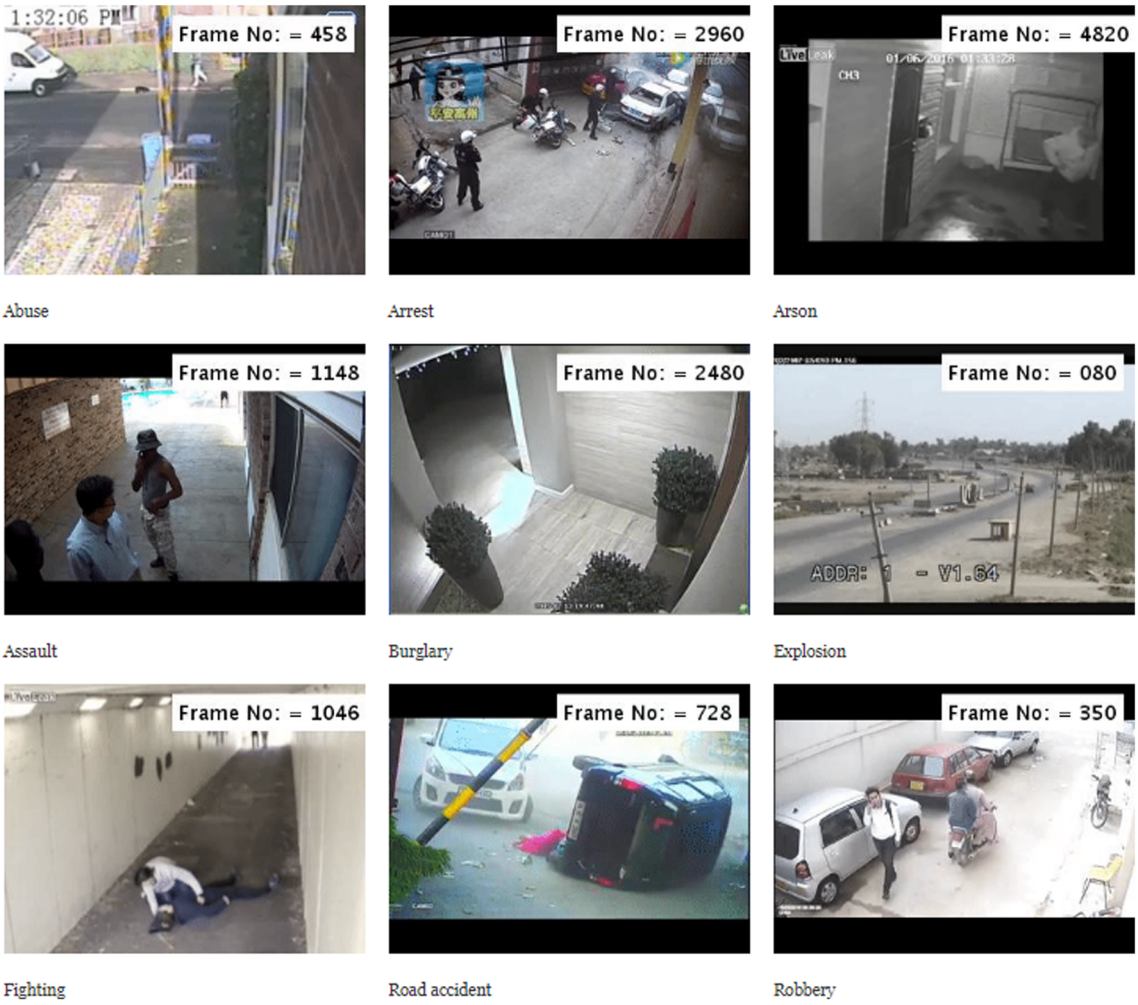


Fig. 2 Sample video dataset

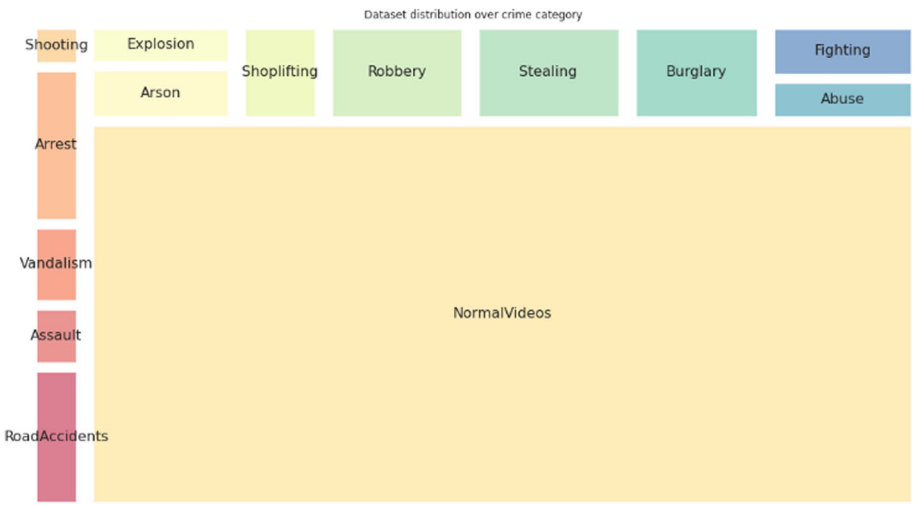


Fig. 3 Dataset distributions over crime categories

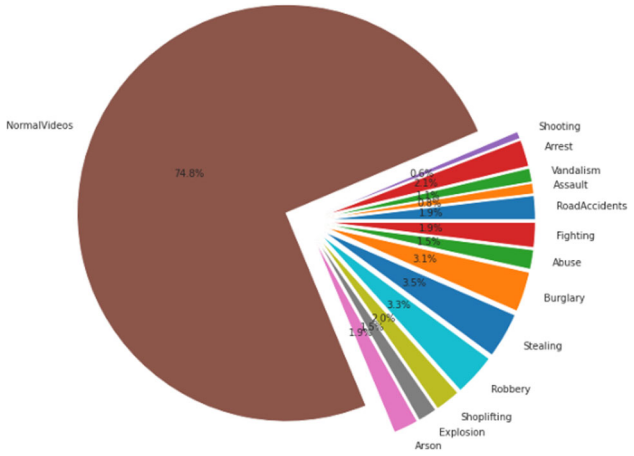


Fig. 4 Ratio of dataset distribution over crime categories

3.3 Feature selection

Feature selection and extraction is the process of identifying the attributes which contributes more information than others in the dataset. These features will be extraction without affecting the distribution of data and without affecting the nature of the dataset. Those selected features will improve the performance of the prediction model. If you consider the video dataset, there may be large number of features like human beings, their actions, location information, background information, other objects in the background, etc. But for this anomaly detection process, we only need objects related to human and their motion. It is an important part in any machine learning model where the data must be reshaped before fed into training phase. This phase also helps in removing the noise if present any. So those features which are not necessary for the anomaly detection model can be removed from the original dataset.

This feature selection methodology will try to find out the correlation between different features and remove those features which have less correlation or no correlation at all. Removing the unwanted features will relatively reduce the dimension of dataset and also reduce the overall variance of the model. The feature selection model identifies some set of features based on their correlation and estimate the performance using the selected features. If they produce a good result, those features will be finalized and passed to the training model. Otherwise, the feature selection method will try to identify another subset. This process will be repeated for much iteration until a desired subset of features was selected.

As the proposed system uses UCF-Crime dataset with video sequences, it is better to use LSTM-CNN technique for feature extraction purpose. The dataset consists of both temporal and spatial information. So the LSTM model which works better for temporal feature selection is combined with the CNN model for extracting spatial information about the data. For improving the feature selection both are combined together to extract maximum trainable parameters from the dataset. The LSTM-CNN technique applies back propagation for improving the feature selection process. The input size to the feature extraction model is defined as 64 X 64 X 3 with a batch size of 16. The number of hidden channels used in this system is 64 and with the kernel size 3X3 in the CNN model. The video dataset is converted into set of frames and fed into the LSTM-CNN model for feature selection.

The features selected for the proposed and existing system are represented in Table 2.

The proposed system selects 5,586,254 features form the input dataset and these features will be further used for object detection and activity classification.

3.4 YOLO object detection

The main objective of the object detection or human detection is to identify the person in the video frames and localize their position. While localizing the human objects, it may become difficult in crowded areas which result in challenging part of localization, since there may multiple objects with frequent movement in a specified area.

In our proposed system, YOLO-based object detection system is used for detecting the trajectory information of the human objects. Unlike other traditional object detection systems, YOLO is faster and achieves frame rate of 45 frames per second. The architecture of YOLO is similar to that of CNN architecture. YOLO takes each frame as an individual image and construct a bounding box for each object in the frame. It converts each image into a 64×64 matrix for creating the bounding box for individual objects and also for the entire frame by globally analyzing the full image. YOLO predicts the objects in the images with bounding box co-ordinates along with their width, height and the confidence score for each frame. The procedure for processing a video sequence YOLO is illustrated in Fig. 5.

An example of human object detection and bounding box creation is shown in Fig. 6.

YOLO is able to predict multiple objects in a frame by creating more bounding boxes in the grid. So this methodology can be useful for tracking human activity in public areas which are crowded. It will provide higher prediction rate than the traditional neural network models. And the performance of YOLO model can be improved by more than 2% with the help of batch normalization technique.

3.5 Activity classification

Once the object is detected and features with higher impact are extracted, the activity recognition process will be initiated. In this proposed work, the YOLO system is improvised

Table 2 Feature selection over different mechanisms

S.No	Model	Dataset	No.of classes Taken	Epochs	Total parameters	Trainable Parameters	Non-trainable parameters
1	2D CNN	UCF—101	10	5	5,963,755	5,962,595	1160
2	VGG16	UCF—Crime	14	5	14,782,158	67,480	14,714,688
3	Proposed System (LSTM—CNN)	UCF—Crime	14	5	7,170,510	5,586,254	1,584,256

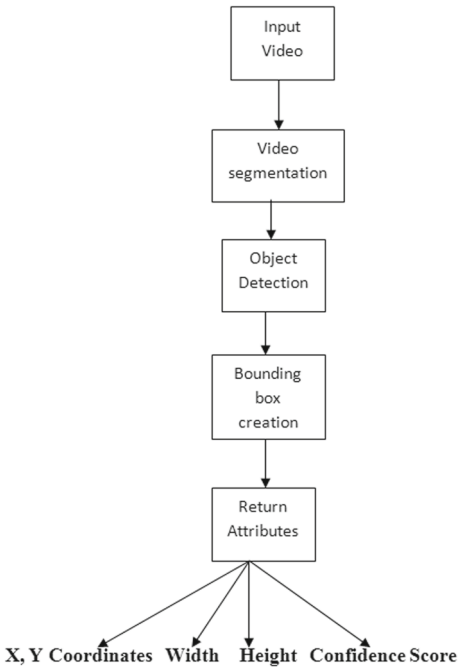


Fig. 5 YOLO object detection Procedure

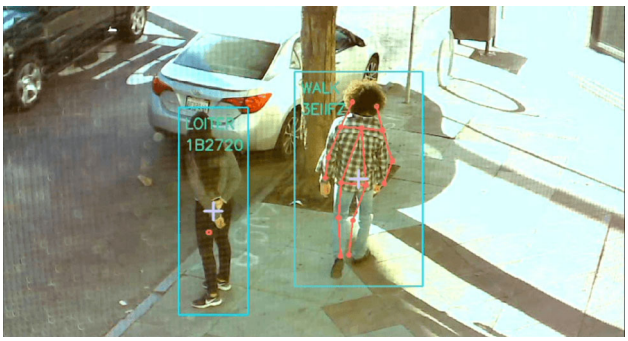


Fig. 6 YOLO human object detection model

by using logistic loss function which computes the probability of each event class with the help of varying probability, the activity can be classified into any one type of crime activity or as a normal activity. The loss function also helpful in suppressing the events with low probabilities and consider the classes with higher probability which in turn identified as the type of event occurring in the place.

The loss function of the YOLO model is defined as follows:

$$\text{CE loss} = \sum_{i=1}^{N} y_{\text{true}}i.\log(y_{\text{pred}}i) \quad (1)$$

where, the N represents the number of categories of events, here the N value is define as 14. Y_true defined as the expected outcome or the type of event, Y-pred defines the predicted event type.

The proposed system resulted in 0.0085 loss during the anomaly activity detection process.

The proposed system is improved with one more feature. Also the hyperparameter tuning has been done by fine-tuning the learning rate. Initially the learning rate was fixed as 0.003. But this learning rate resulted in faster learning where the classification results are not satisfactory. So the learning rate was gradually reduced for encouraging slower learning process as the dataset is more complex with more features. After much iteration the system achieved 0.91 AUC at the learning rate of 0.00003.which is kept as the optimal learning rate for the proposed system.

As the system is completely trained with all the fourteen types of crime activities this system is able to detect the crime activities from any custom data received form the camera. The YOLO model is integrated with the real time alarm system for alerting the surrounding in case of any abnormal activity takes place.

3.6 Alerting mechanism

Once the abnormal activity is detected by the system, it sends an alerting signal to the sound alarm attached to the system located in the place of crime. The alert system uses a sound alarm with 120 dB, which will ring for every 60 s until the authority turns it off. Thus the alarm will alert the nearby people about the occurrence of crime event and prevent further precedence of the crime activities. This can also be helpful in taking immediate countermeasures in the place of crime.

4 Experimental results

The proposed system is implemented with LSTM-CNN algorithm for feature selection and improvised YOLO algorithm for activity recognition and classification. the system is trained with UCF-Crime dataset which defines 14 categories of abnormal crime activities. The dataset is given as video sequence input to the system with 64 × 64 size and batch size of 64 for 4 epochs. The results from the proposed system are compared with existing system for the same UCF-Crime dataset.

The performance of the proposed system is evaluated with the following parameters. In deep learning techniques achieving better accuracy and providing a better fit algorithm is a crucial task.

4.1 Confusion matrix

The confusion matrix provides a way to understand how well the predictions have been done by the model. It returns four types of values such as TP (true positive) which defines the correctly predicted values, TN (true negative) which defines the values which does not belong to the classes correctly, FP (false positive) that defines a value that is mis-predicted as a positive value and FN (false negative) which denotes a value mis-predicted as negative.

4.2 Precision

Precision is defined as how much correct predictions of the proposed model are actually true. It calculates the ratio as follows:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \tag{2}$$

A model with no false-positive prediction produces the precision as 1.

4.3 Recall

Recall also verifies how accurate the model predictions are made, through true positive and false negative.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \tag{3}$$

Suppose the model does not have any false-negative values, then it is said to be have recall as 1.

4.4 F1-score

The F1—score provides a trade-off between the precision and recall. The outcome of the performance metrics like precision, recall, F1-score and support is represented in Fig. 7

	precision	recall	f1-score	support
0.0	0.00000000000000	0.00000000000000	0.00000000000000	297
1.0	0.02922422954304	0.01634472511144	0.02096436058700	3365
2.0	0.14473456683382	0.29967776584318	0.19519589552239	2793
3.0	0.00000000000000	0.00000000000000	0.00000000000000	2657
4.0	0.03177570093458	0.01332114405119	0.01877243029355	7657
5.0	0.15202444614209	0.03056835637481	0.05090164982734	6510
6.0	0.00289136720363	0.00568643379366	0.00383351588171	1231
7.0	0.71411345807385	0.77503387116640	0.74332754992802	64952
8.0	0.11628348850466	0.22981599699587	0.15442846328539	2663
9.0	0.00013559322034	0.00119760479042	0.00024360535932	835
10.0	0.03797468354430	0.00157273918742	0.00302038761641	7630
11.0	0.36850649350649	0.05955660501115	0.10254093732355	7623
12.0	0.07916241062308	0.23437500000000	0.11835072537541	1984
13.0	0.01370757180157	0.01890189018902	0.01589103291714	1111
accuracy			0.47709957954505	111308
macro avg	0.12075242928082	0.12043229517961	0.10196218242266	111308
weighted avg	0.46450534360943	0.47709957954505	0.45679368812060	111308

Fig. 7 Perfromance metrics

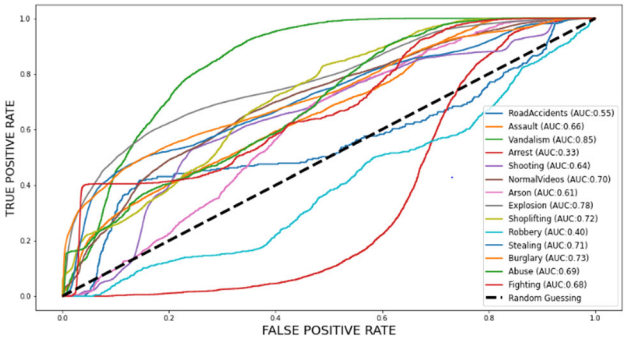


Fig. 8 AUC curve with learning rate 0.003

4.5 ROC curve

The ROC (receiver operating characteristics curve) provides a graphical representation for analyzing the performance of deep learning model in terms of TPRate and FPRate.

The true-positive rate is calculated as follows:

$$\text{TPRate} = \text{TP} / (\text{TP} + \text{FN}) \tag{4}$$

$$\text{FPRate} = \text{FP} / (\text{FP} + \text{TN}) \tag{5}$$

For evaluating the models in a better manner, the TPRate and FPRate will be calculated with different thresholds for the classification model.

4.6 AUC curve

The next performance measure is AUC (area under the ROC Curve). It provides a region that is represented under ROC curve from (0,0) to (1,1). AUC can be calculated to measure the performance of the model irrespective of the threshold. The accuracy over AUC is mentioned in Figs. 8 and 9

At the maximum of 91% accuracy has been achieved for predicting vandalism activities. When the learning rate is

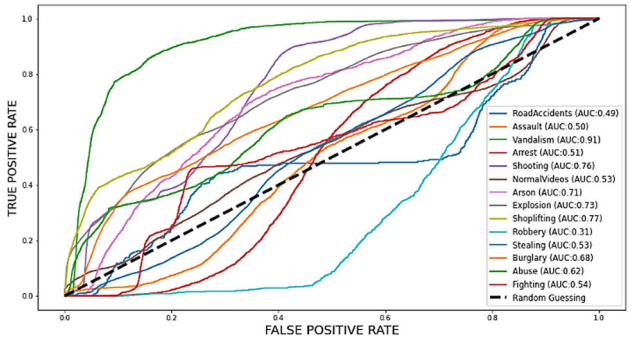


Fig. 9 AUC curve with learning rate 0.00003

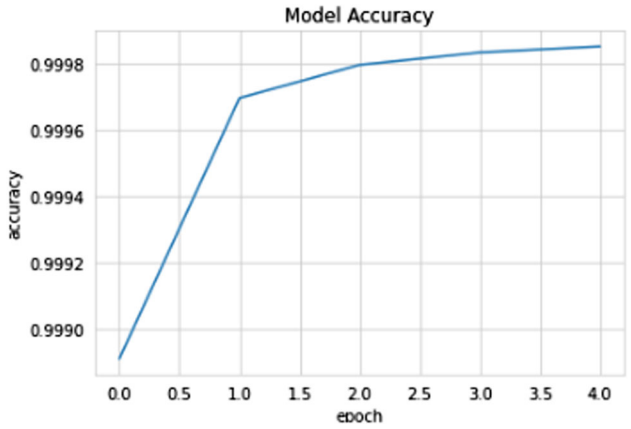


Fig. 10 Model accuracy

fine-tuned, it collectively achieves an accuracy of 82.99 over all the 14 classes.

4.7 Accuracy

The accuracy of the model is defines as the ratio between the correctly made predictions and total number of predictions. It can be defined as follows:

$$\text{Ac} = \text{TP} + \text{TN} / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (6)$$

A model achieves higher accuracy when it makes correct predictions about both positive and negative class. The proposed system achieves a trainining accuracy of 99.89 and validation accuracy of 99.96 after completing five epochs. Figure 10 depicts the accuracy of the proposed model.

4.8 Categorical entropy loss

The Softmax cross-entropy loss has been used in the proposed system for multiclass crime classification. Loss is generally defined as the difference between the actual and predicted outcomes. Since the proposed model consists of 14 classes, we are using multiclass categorical cross-entropy (CCE) for determining the loss.

for each_element in batch:
count non-zero labels.
for class in num_class:
if (labels belongs to class)
*logprobs(element) + = - np.log(probability (element, class)) * label (element , class)*
loss = sum(logprobs)

Algorithm

The calculation of cross-entropy represented as follows:

$$\text{CE loss} = \sum_{(i=1)}^{N} y_{\text{true}} i \cdot \log(y_{\text{pred}} i)$$

The loss function for the proposed and existing is plotted in Table 3.

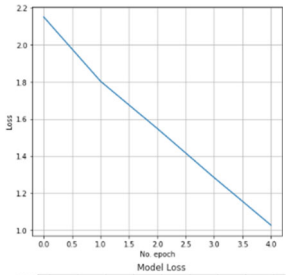
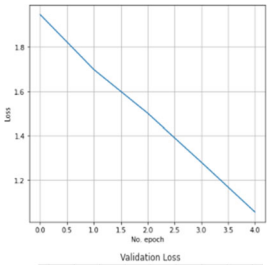
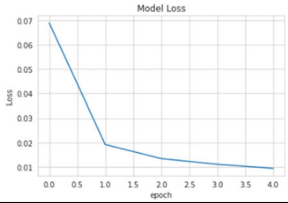
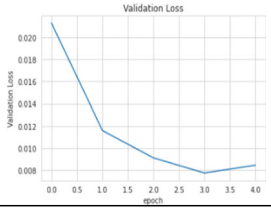
5 Conclusion

The anomaly detection system proposed in this paper uses a lightweight machine learning algorithm improvised YOLO for human activity detection and classifying the human activities. This system is able to detect 14 classes of crime activities with more accuracy than the existing models and also selects more features for training the system with the proposed LSTM-CNN model. The proposed algorithm provides better performance than the existing ones by achieving relatively higher AUC of 91% for detecting the abnormal event category of vandalism after fine-tuning the learning rate with 0.0085 losses at the end of four epochs. It nearly took 4.5 h for the execution of the model. The system detects the abnormal activities like detecting crime activities like abuse, arrest, arson, assault, road accident, burglary, explosion, fighting, robbery, shooting, stealing, shoplifting, and vandalism. The proposed system is able to distinguish the normal events from the 14 classes of crime activities as shown in Fig. 9 of AUC curve performance. So if this system is deployed in public areas or in private organizations, this can provide a secure environment and ensure the safety of public.

6 Future work

For the future work, the system may be extended by adapting a transfer learning model for better feature selection and may be tested with higher number of epoch. The feature extraction mechanism may be improved to select more features than the proposed system and reduce the false positive rate. Also

Table 3 Loss comparison

Model	Training loss	Validation loss
CNN		
Proposed system		

the system may be designed to select more domain specific features for better prediction in the field of crime detection and women safety.

Author contributions KG and VS contributed to conceptualization and methodology; KG contributed to software, resources, writing—original draft preparation, visualization, writing—review and editing and validation; SV contributed to formal analysis and investigation, supervision, and project administration.

Data availability Data used in the paper available as video dataset at https://www.crcv.ucf.edu/projects/real-world/ used for training and testing purpose in the paper.

Declarations

Conflict of interest In accordance with publisher and my ethical obligation as a researcher, I am reporting that there are no relevant financial or non-financial competing interests to report.

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